A TALK ABOUT THE MACHINES THAT ARE ABOUT TO MOVE IN

Who governs the robots?

What we found when we read every rule on Earth for machines that think and move

Somewhere, right now, a robot is being built to live in your mother's room.

Not a factory arm behind a fence. A machine that lifts, carries, reminds, watches, and talks back.

SO WE WENT LOOKING FOR THE RULES

3,320

documents: laws, standards, strategies, studies, news

119

countries, plus the EU and the UN system

0

humans who read them all. An AI did, 39 questions per document, and other AIs checked its work.

278

binding rules on Earth that reach a robot.

Laws with teeth: break them and something happens to you. Each counted once, paperwork excluded.

25

of them mention artificial intelligence.

The other 253 were written for aircraft, vehicles, machinery, products and medical devices. Robots are governed sideways.

GOVERNED SIDEWAYS

146 rules for drones. A drone is an aircraft.

109 rules for self-driving cars. A car is a vehicle.

13 rules for factory robot arms. An arm is a machine.

AND THE ROBOTS WE WILL LIVE WITH?

0

binding rules for companion robots

0

for humanoids

4

for care robots, all of them laws about something else that mention care robots in passing

Among 429 documents about companion, care and humanoid robots. The one text written for care robots is a voluntary standard from 2014.

COMPANION ROBOTS

43%

of the research papers are about them

3%

of the rules are

Scientists study the robots we will live with. Lawmakers regulate the ones already in the street.

Rules follow machines into public space. Usually after the first accident.

Drones
Rules mostly 2016 to 2021, after the near-misses at airports

Self-driving cars
Rules from 2017, after the test fleets, and the first deaths

Sidewalk robots
State laws 2018 to 2022, after the robots were already on the pavement

Three quarters of the 278 rules date from 2019 or later; 105 of them from 2024 to 2026.

WHEN THE INTELLIGENCE HAS A BODY, THE QUESTIONS CHANGE

Can it hurt you?

How close may it come?

Who pays when it breaks something?

How much may it decide alone?

Physical safety is spelled out in 167 of the 278 rules. Fairness in 3. Human dignity in 1. The rulebook for a body is about the body.

THE HARDEST QUESTION: HOW MUCH MAY IT DECIDE ALONE?

171

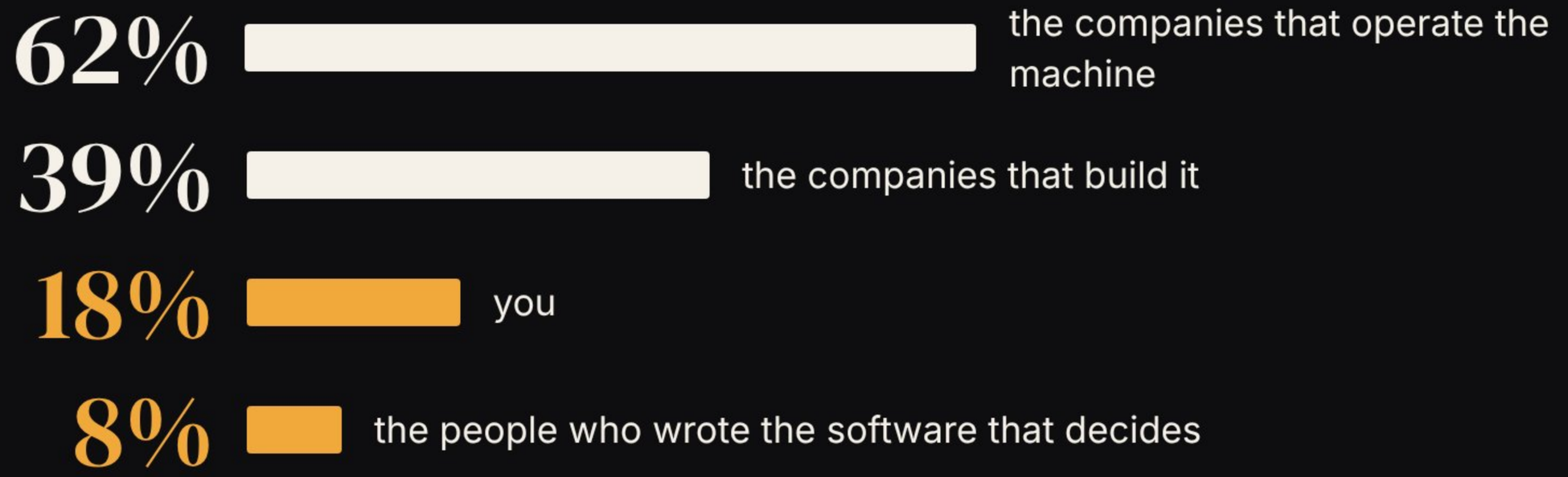
rules raise it

73

rules answer it: what is allowed,
by whom, checked how

Mention is not detail.

WHO HAS TO OBEY THESE RULES?



Share of the 278 binding rules that name each party in their own text; a rule can name several.

We used AI to read the laws about AI. Here is how much to trust that.

73%

of a random hundred rules survived an independent re-read on three strict tests. Up from 54% a year ago. Not 100.

0

documents labelled by a human so far. A map drawn by machines, waiting for a person to walk it.

THREE THINGS, BEFORE THE ROBOTS ARRIVE

1 Write the rules first this time.

Companions, carers, humanoids: zero, four, zero. We can see them coming. Drones did not get that courtesy.

2 Answer the control question. Do not just raise it.

Who may let the machine decide what, under which conditions, checked by whom. Aviation did it. Everyone else can.

3 Make the safety knowledge public.

The documents that say how close, how fast, how much force are standards, and 57 of our 61 are behind a paywall. Rules you cannot read are rules you cannot follow.

The question is not whether robots will come into our homes.

It is whether the rules will be there when they do.

Data, code and every number: github.com/cjimmylin/embodied-ai-governance · Explore the 3,320 documents: claude.ai/artifact/YKsXJRaT7tSfRXseuo5kuZ